

Supporting Information

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SI Materials and Methods

Animals. Cortical hemispheres from four baboons were used in this study. The brain of case 09-27 (*Papio cynocephalus/Anubis cross*) was purchased from the University of Washington National Primate Research Center. Cases 11-31, 10-04, and 11-45 were collected from baboons euthanized at the Texas Biomedical Research Institute as approved by its Institutional Animal Care and Use Committee. Table S1 summarizes specific details on the age, sex, size, and perfusion method in each case.

Flattened Cortex Dissection and Sample Preparation. The flattened cortices were fixed with 2–4% paraformaldehyde (PFA; 20–40 g/L) for a minimum of 2 wk before dissection. Postmortem degradation of proteins in unperfused brains that remain in the head generally starts to become evident at 24 h postmortem (1), and our flattened cortices were fixed with PFA well within this time frame.

For dissection, each cortical hemisphere was photographed and viewed on a light box. Readily identifiable cortical areas were dissected from the cortical sheet with a scalpel, including the primary visual cortex (V1), primary sensory cortex (S1), and primary motor cortex (M1). These areas are readily identified when the flat cortex is viewed on a light box to visualize myelinated sensory areas that appear darker relative to surrounding cortical areas, and it has been found to provide a reliable estimate of areal boundaries (2–4). In cases where the cortical hemisphere remained intact, readily identifiable cortical areas were dissected from the remaining cortex using intact sulcal landmarks. Once removed, V1, S1, and M1 were further dissected into smaller pieces that measured ~5 mm × 5 mm, or smaller, and each piece was assigned to an internal representation (example, M1 face representation; M1 forelimb representation, and so forth) when possible. The remaining cortex was further dissected into small pieces that measured ~5 mm × 5 mm. The exact location of dissection cuts was drawn onto a high-resolution photograph of the cortical image. Each piece was assigned a number and weighed. The surface area was measured for each piece using ImageJ software (National Institutes of Health). This software was also used to assign an anterior-posterior coordinate by generating a centroid measure for the location of each tissue piece. Each cortical piece was processed for cell and neuron density as one sample.

Each cortical tissue piece was homogenized using a glass Tenbroeck tissue grinder (Fisher Scientific) in a dissociation solution of sodium citrate and triton X-100 in distilled water. The homogenized suspensions contained free-floating nuclei. Each suspension sample was brought to a specific volume, by adding PBS to the sample, and was determined based on suspension density. In this study, suspension volumes from each sample were in the range of 2–6 mL.

Cell Density Estimates. The isotropic fractionator method (5) and flow fractionator method (6) were used to determine cell density. There is excellent agreement in cell estimates in samples where counts were obtained using both methods, and we can confidently compare the data from cases using either method. Data from case 10-04 and case 11-31 were used to develop the flow fractionator cell-counting method (6), which demonstrates not only the excellent agreement between methods in cases relevant to this study, but also shows that the cell estimates obtained by both cell-counting methods works in both normal and pathological tissue.

Main suspension samples from cases 09-27, 10-04, and 11-45 were individually well mixed to ensure uniform density of nuclei in suspension, and a 0.5-mL aliquot was sampled to obtain nuclei density measure. The 0.5-mL aliquot was centrifuged and 80 μ L of solution was removed and replaced with DAPI (10 mg/L). DAPI binds to DNA and therefore labels all nuclei in the sample, regardless of cell type, and fluoresces with an UV excitation (460-nm emission). The samples were well mixed and DAPI-stained nuclei were counted to estimate the total number of cells using fluorescence microscopy and a Neubauer counting chamber with glass coverslip. Nuclear density for the original samples were determined by multiplying the mean nuclear density obtained with the counting chamber by the total suspension volume to give the total number of nuclei in the sample.

Main suspension samples from case 11-31 were thoroughly mixed and a 50- μ L aliquot was added to a 250 μ L of DAPI (2 mg/L). A fixed volume of 50 μ L of Countbright absolute counting beads (Invitrogen) was added to each sample. The nuclei density of each sample was estimated using a Becton Dickinson (BD) 5-laser LSR II flow cytometer equipped with 355-, 405-, 488-, 532-, and 635-nm lasers. The 355-nm laser and 450/50 bandpass filter were used to evaluate the number of DAPI⁺ nuclei events that occurred with 1,000 bead events. BD FACSDiva v6.1.3 software was used to analyze the data. Two-parameter interrogation of Side Scatter (SSC-A) and DAPI (DAPI-A) was used to place a selection gate around DAPI⁺ nuclei for quantification. As many nuclei as possible were included in the gate, although avoiding small debris in the samples and including multiplet data (6). The nuclei in each gate were summed and the nuclear density was calculated for the original sample according to the guidelines for Countbright beads (Invitrogen). The number of nuclei was determined for each sample by calculating the nuclei per microliter by multiplying the number of cell events per the number of bead events and the assigned bead count per volume of the sample aliquot (50 μ L). The total sample suspension volume was multiplied by the nuclei concentration to determine the number of nuclei in the total sample, thereby estimating the total number of cells in the sample. All samples were prepared in duplicate to detect sample preparation variation and confirm accuracy of nuclear estimates. All flow-cytometry experiments were conducted at the Vanderbilt Medical Center Flow Cytometry Shared Resource.

Neuron Density Estimates. Neuronal fraction for all cases was determined using the flow fractionator method (7). Each sample suspension was thoroughly mixed and a 0.5-mL aliquot was sampled to label neuronal nuclei using immunocytochemical techniques. All samples were antigen-retrieved for 40 min in 0.2 M boric acid in an oven set at 75 °C, washed once in PBS, and resuspended in PBS and the anti-NeuN antibody (Millipore) overnight. The following day, samples were washed three times in PBS and resuspended in Alexa Fluor 674 goat anti-mouse IgG secondary antibody (AF647; Invitrogen). DAPI was also added to concurrently label all cell nuclei. Samples were washed three more times in PBS after 1 h of incubation, and resuspended in 0.5 mL PBS. All samples were forced through a 35- μ m filter to remove debris from the sample. The proportion of nuclei in the sample that was neuronal was quantified using the BD LSR II. The AF647 secondary antibody is excited by the 635-nm laser and its emission is captured by a 680-nm dichroic long-pass mirror and a 720/20 bandpass filter. For all samples, 5,000–10,000 DAPI⁺ cells were evaluated for positive AF647 (AF647⁺) label. The distributions of forward and side light scatter pro-

duced by each nucleus as it passed through the laser were used to place a selection gate around the nuclei to quantify the proportion of AF647⁺ nuclei in the total number of DAPI⁺ events. To determine the number of neurons in a sample, the total number of nuclei in the sample was multiplied by the percentage of neuronal nuclei.

Histology, Light Microscopy, and Verification of NeuN-Immunoreactivity.

The intact right hemisphere of case 11-45 and intact left hemisphere of case 11-31 was sectioned into 50- μ m-thick frozen sections into 10 series. One series from each hemisphere was processed for Nissl substance (with thionin), and another series was labeled for NeuN protein using a commercial antibody against NeuN (monoclonal anti-NeuN antibody, Millipore; 1:5,000) to verify tissue and staining quality in this study. The primary antibodies were labeled with biotinylated secondary antibodies (Vector Labs) amplified using an avidin/biotin conjugate kit (Vector Labs) and visualized using a diaminidobenzidine reaction with nickel enhancement. Digital photomicrographs of the cortex were captured using a Nikon DXM2200 camera (Nikon) mounted on a Nikon E800 microscope. The contrast was adjusted on the final images using Adobe Photoshop (Adobe Systems), but images were otherwise unaltered. Balaram et al. (8) have previously described the detailed immunohistochemical procedures used in this study.

We confirmed that the anti-NeuN immunolabel correctly labeled neurons in tissue from our normal and epileptic baboons. Stained sections from the prefrontal cortex (Brodmann's area 46) in the intact hemispheres of normal (case 11-31, left hemisphere) and epileptic (case 11-45, right hemisphere) baboons indicate that the NeuN stain is effective in staining neurons in the frontal lobe, which was the region that was most affected by neuron reduction in the epileptic cases. The Nissl- (Fig. S2A) and NeuN-

(Fig. S2B) stained sections showed that the tissue preparation and condition was excellent in both cases. The presence of background or staining artifact was minimal-to-none. The laminar organization of cortical layers in this area appears normal in both cases. High-magnification images of NeuN labeling in cortical layer 5 of normal and epileptic (Fig. S2B) baboons show consistent and excellent staining of pyramidal and other neurons. The neuron reduction that was revealed in neuron counts is not visually obvious in this material, but was also not closely examined for laminar differences at this time. We also visually confirmed labeling of neuronal nuclei in homogenized suspensions using the NeuN antibody. Fig. 1C shows NeuN-immunoreactive (NeuN-IR) nuclei in homogenized suspension samples from the frontal lobe of cases 11-31 (right hemisphere) and 11-45 (left hemisphere) visualized with Alexa Fluor 594 goat anti-mouse IgG secondary antibody (AF594; Invitrogen). Nuclei that were NeuN-IR were brightly labeled, and there was minimal-to-no background staining that would interfere with determining labeled nuclei from background. The lack of background or artifact staining demonstrates that the sample homogenization process did not result in broken or damaged nuclei and that the anti-NeuN antibody was specific to NeuN-IR nuclei.

Data Analysis and Summary. The overall number of cells and neurons were compared for epileptic and neurologically normal conditions using a Mann–Whitney test. Comparisons were also made between V1, S1, and M1 between the two conditions using the Mann–Whitney test. Statistical significance was set at $P < 0.05$, and was analyzed using SPSS. Table S2 is provided to summarize the data. The total numbers of cell and neurons in the cortex are summarized for each case. Total numbers of cells and neurons in primary visual cortex, primary somatosensory cortex, and primary motor cortex are also reported in Table S2.

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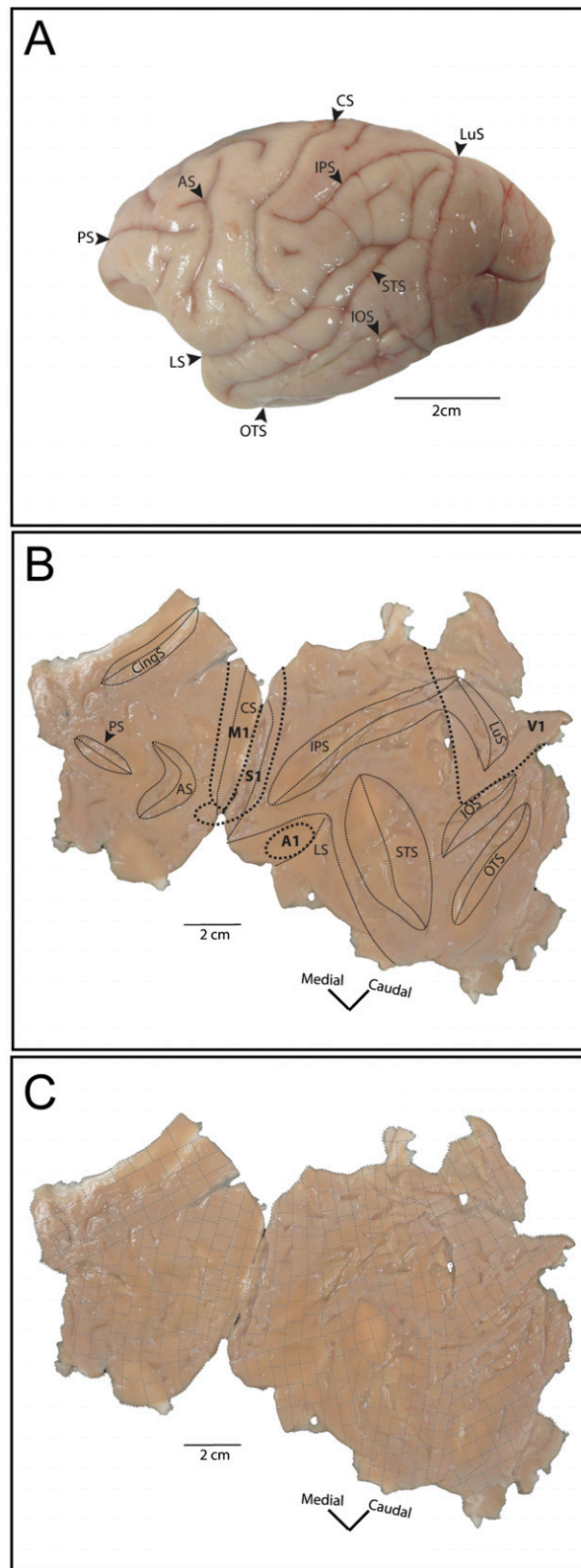


Fig. S1. (A) A lateral view of the intact left hemisphere of a baboon brain (case 11-45). The locations of major sulci are indicated. (B) Left hemisphere following manual flattening. The locations of opened major sulci are indicated. The relative locations of primary cortical areas are also indicated on the flattened cortical image. (C) The flattened left hemisphere shown in *B* cut into 428 tissue pieces. Identifiable cortical areas were first dissected from the remaining cortex to protect the integrity of the cortical borders, and then all tissue was dissected into $\sim 5 \times 5$ -mm tissue pieces. Actual surface areas of each piece were later measured with ImageJ software on high-resolution photographs of the flattened, dissected cortical hemispheres. A1, primary auditory cortex; AS, arcuate sulcus; CingS, cingulate sulcus; CS, central sulcus; IOS, inferior temporal sulcus; IPS, intraparietal sulcus; LS, lateral sulcus; LuS, lunate sulcus; M1, primary motor cortex; OTS, occipitotemporal sulcus; PS, principal sulcus; S1, primary somatosensory cortex; STS, superior temporal sulcus; V1, primary visual cortex.

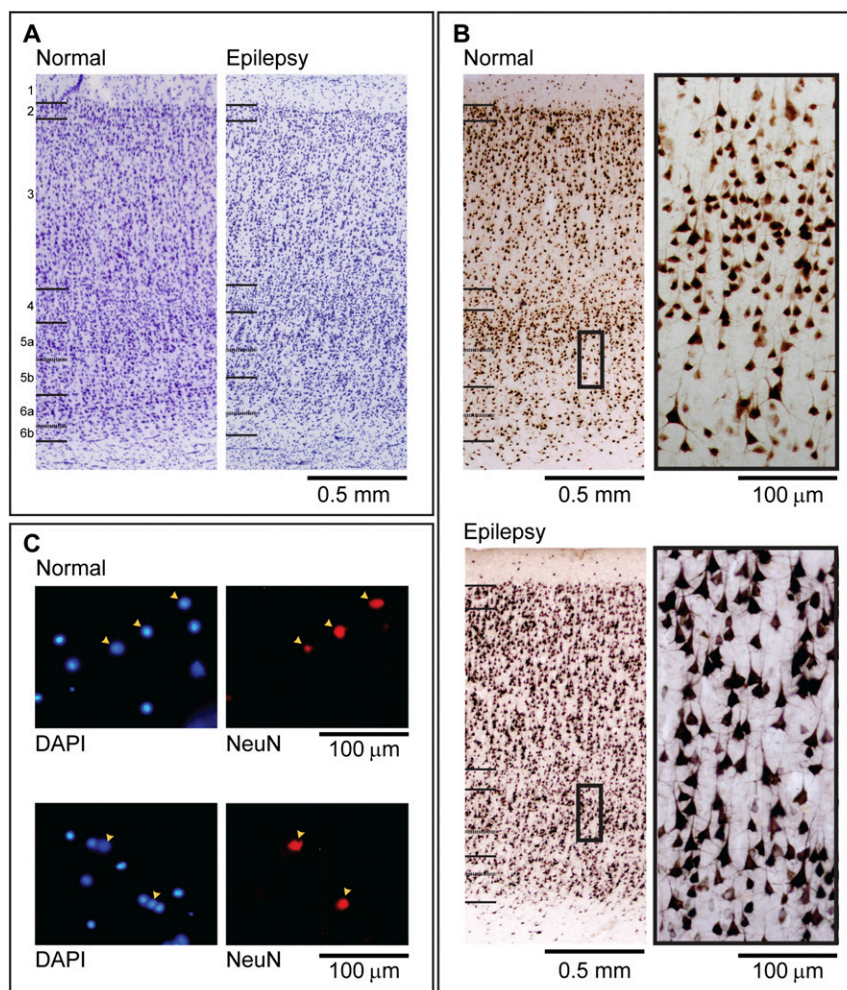


Fig. S2. High-magnification images of the laminar organization of ventral area 46 in the frontal cortex of a normal baboon (case 11-31) and an epileptic baboon (case 11-45). Sections were stained with Nissl (A) and NeuN (B) to verify the quality of tissue preparation and consistent NeuN labeling, particularly in the frontal cortex where cell counts were found to be lowest. High-magnification image of layer 5 NeuN-IR cell bodies shown in B further confirms consistent and accurate NeuN labeling in the frontal cortex in a baboon with seizures. In homogenized samples from the frontal cortex in opposite hemisphere for each case, nuclei were shown to robustly label DAPI and NeuN with minimal background (C).

Table S1. Summary of experimental cases

Case #	Species	Age (y)	Sex	Perfusion method	Weight		Neurological condition	Hemisphere	Cortical surface area (cm ²)
					Body (kg)	Brain (g)			
09-27	PHX	12.6	Female	0.9% PBS + 4% PFA (postfixed)	17.8	173.4	Normal	Right	186
11-31	PHA	13	Female	0.9% PBS + 4% PFA (postfixed)	21.7	N/A	Normal	Right	234
10-04	PHA	14	Female	0.9% PBS + 4% PFA (postfixed)	18.6	165.3	Epilepsy	Right	242
11-45	PHX	17	Female	0.9% PBS + 2% PFA	23.4	159.7	Epilepsy	Left	192

PHA, *Papio hamadryas anubis*; PHX, *Papio hamadryas anubis/cyncephalus* hybrid.

Table S2. Summary of cell and neuron data

Case #	Neurological condition	Total # cells	Cell density (cells/cm ²)	Total # neurons	Neuron density	Average % neurons
Summary of data averaged across all areas and regions of cortex						
09-27	Normal	4.67 billion	25.2 million	2.39 billion	12.9 million	50
11-31	Normal	4.29 billion	18.6 million	2.26 billion	9.77 million	51
10-04	Epilepsy	4.00 billion	16.5 million	1.61 billion	6.59 million	39
11-45	Epilepsy	4.24 billion	22.7 million	1.79 billion	10.0 million	41
Summary of data averaged across V1						
09-27	Normal	878 million	29.5 million	652 million	22.0 million	75
11-31	Normal	804 million	24.9 million	587 million	17.9 million	72
10-04	Epilepsy	519 million	19.4 million	342 million	12.8 million	66
11-45	Epilepsy	727 million	34.4 million	515 million	24.2 million	71
Summary of data averaged across S1						
09-27	Normal	74.5 million	22.3 million	41.1 million	11.9 million	55
11-31	Normal	80.9 million	17.3 million	48.2 million	10.2 million	59
10-04	Epilepsy	57.5 million	10.3 million	15.5 million	2.82 million	26
11-45	Epilepsy	41.2 million	20.0 million	16.2 million	7.83 million	39
Summary of data averaged across M1						
09-27	Normal	153 million	22.7 million	55.8 million	8.25 million	36
11-31	Normal	152 million	23.9 million	62.9 million	9.57 million	40
10-04	Epilepsy	149 million	13.9 million	28.1 million	2.61 million	18
11-45	Epilepsy	81.7 million	16.2 million	19.2 million	3.81 million	24